

ENGG1910: Demystifying AI

Lab 4 Guidelines

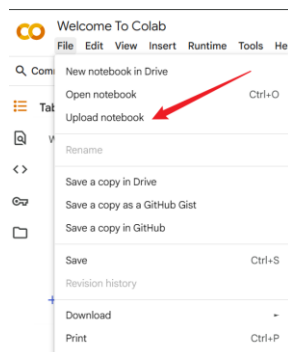
1 Objectives

1. Try to clone your own voice.
2. In this lab, students are expected to understand the basic procedure of **Text-to-Speech**, including **data preparation**, **feature extraction**, **model training** and **model inference**.
3. Create different prompts to test the provided **VALL-E X** model. Observe and compare the generated results and make some conclusion.

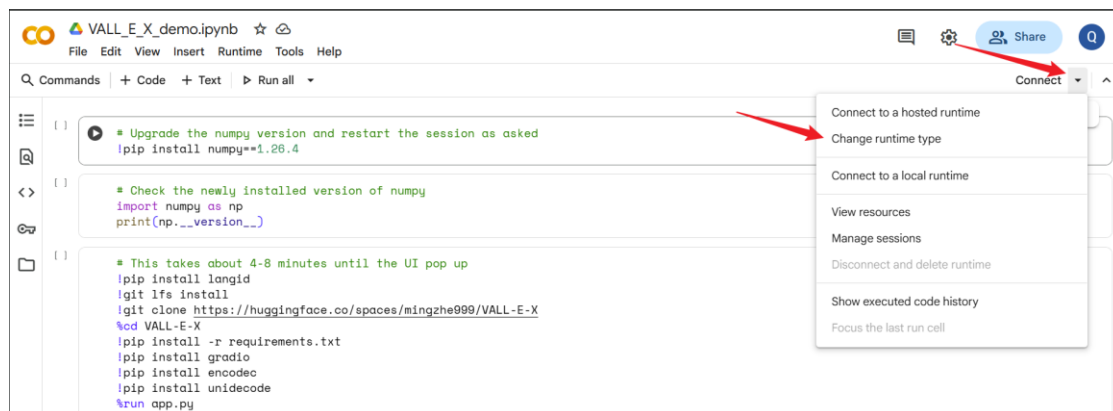
2 Guidance for Objective 3

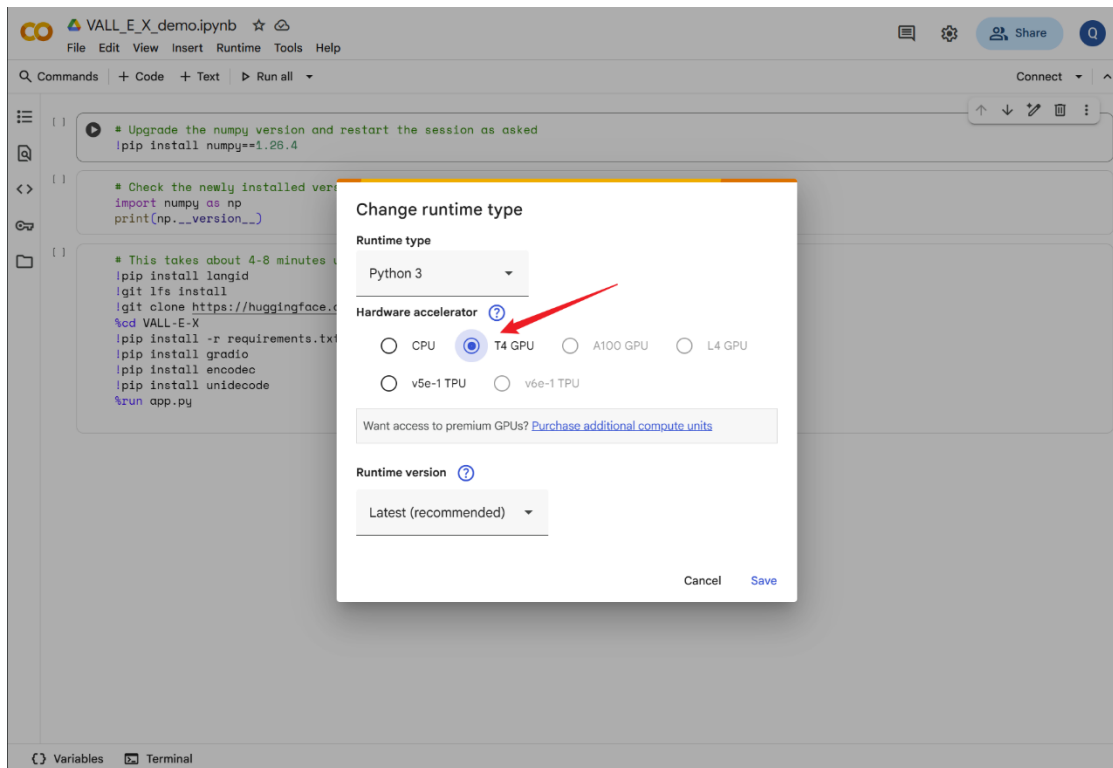
Notice: To ensure recording quality, if you have headphones (with microphone), please bring them to the class if possible. We will also provide 1-2 sets of devices for use.

2.1 Upload the code file VALL_E_X_demo.ipynb to <https://colab.research.google.com>

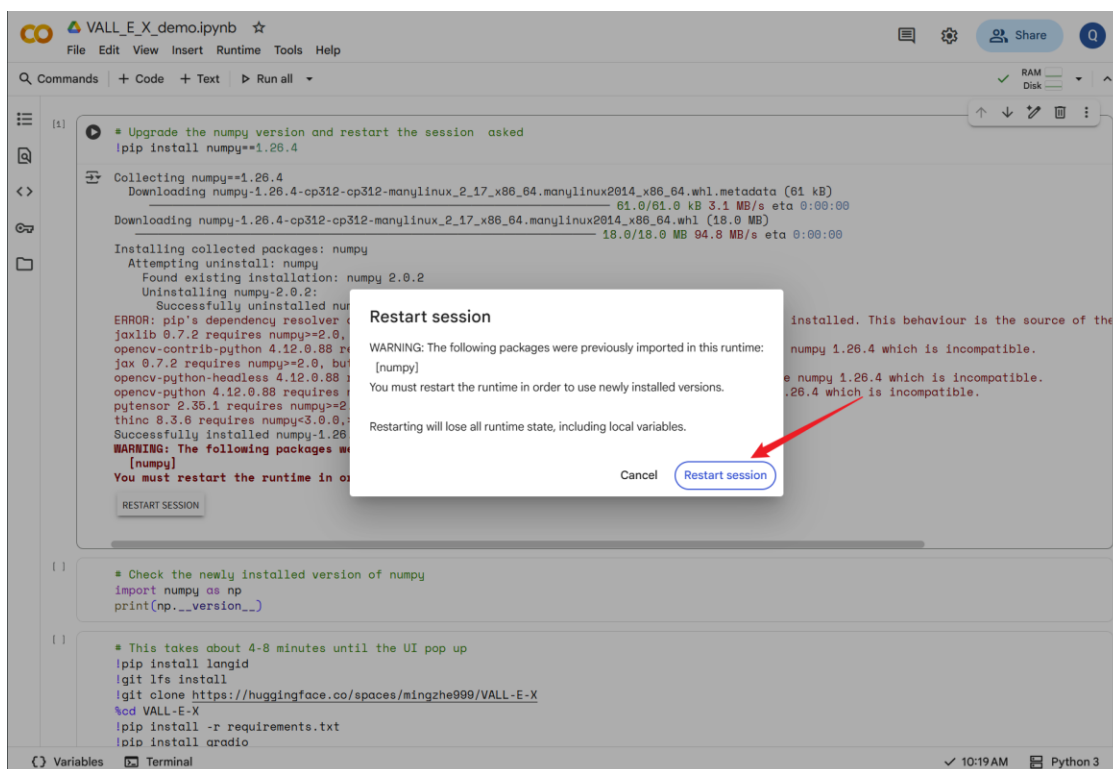


2.2 Change the runtime type into T4GPU to enhance the speed





2.3 Run the **first** code cell and restart the session if asked. Then run the **second** code cell to check if we successfully upgrade the package.



```
[1] ✓ 0s
# Check the newly installed version of numpy
import numpy as np
print(np.__version__)

1.26.4
```

2.4 Run the last cell and wait for 4-8 minutes until the UI pops up. You will get your own link!

special_tokens_map.json: 2.19k/? [00:00<00:00, 121kB/s]

Special tokens have been added in the vocabulary, make sure the associated word embeddings are fine-tuned or trained.

config.json: 1.99k/? [00:00<00:00, 240kB/s]

model.safetensors: 100% 3.06G/3.06G [02:07<00:00, 35.0MB/s]

generation_config.json: 3.75k/? [00:00<00:00, 271kB/s]

[nlk_data] Downloading package punkt to /root/nltk_data...

[nlk_data] Unzipping tokenizers/punkt.zip.

It looks like you are running Gradio on a hosted Jupyter notebook, which requires 'share=True'. Automatically setting 'share=True' (you can

Colab notebook detected. **To show errors in colab notebook, set debug=True in launch()**

* Running on public URL: <https://df2c8f3...@gradio.live> **Your own public URL!**

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run 'gradio deploy' from the terminal in the working directo

VALL-E X

[Duplicate Space](#) or [Open in Colab](#) to skip the queue.

VALL-E X can synthesize high-quality personalized speech with only a 3-second enrolled recording of an unseen speaker as an acoustic prompt, even in another language for a monolingual speaker.

This implementation supports zero-shot, mono-lingual/cross-lingual text-to-speech functionality of three languages (English, Chinese, Japanese)

See this [demo](#) page for more details.

[Infer from audio](#) [Make prompt](#) [Infer from prompt](#) [Infer long text](#)

Upload a speech of 3~10 seconds as the audio prompt and type in the text you'd like to synthesize.

The model will synthesize speech of given text with the same voice of your audio prompt.

The model also tends to preserve the emotion & acoustic environment of your given speech.

For faster inference, please use "Make prompt" to get a .npz file as the encoded audio prompt, and use it by "Infer from prompt"

Text Message

Next steps: [Deploy to Cloud Run](#)

2.5 Go to your public URL

VALL-E X

[Duplicate Space](#) or [Open in Colab](#) to skip the queue.

VALL-E X can synthesize high-quality personalized speech with only a 3-second enrolled recording of an unseen speaker as an acoustic prompt, even in another language for a monolingual speaker. This implementation supports zero-shot, mono-lingual/cross-lingual text-to-speech functionality of three languages (English, Chinese, Japanese)

See this [demo](#) page for more details.

[Infer from audio](#) Make prompt Infer from prompt Infer long text

Upload a speech of 3~10 seconds as the audio prompt and type in the text you'd like to synthesize.
The model will synthesize speech of given text with the same voice of your audio prompt.
The model also tends to preserve the emotion & acoustic environment of your given speech.
For faster inference, please use "Make prompt" to get a `.npz` file as the encoded audio prompt, and use it by "Infer from prompt"

Text

Welcome back, Master. What can I do for you today?

language

auto-detect

accent

no-accent

Transcript

Write transcript here. (leave empty to use whisper)

Message

Output Audio

Generate!

Prompt name

prompt_1

Make prompt!

2.6 Upload audio prompt

uploaded audio prompt

Drop Audio Here
- or -
Click to Upload

recorded audio prompt

Record from microphone

Two way to upload your recording:

- a. If you have already prepared your recording file, you can upload it directly.

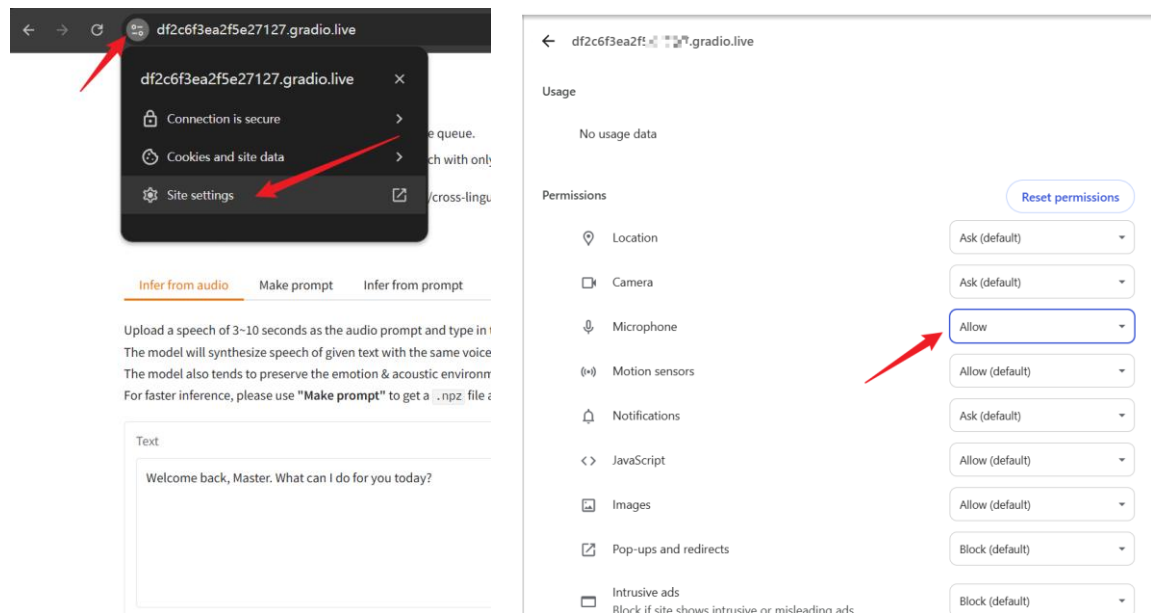
Drop Audio Here

- or -

Click to Upload

- b. Record your voice through the microphone of your device.

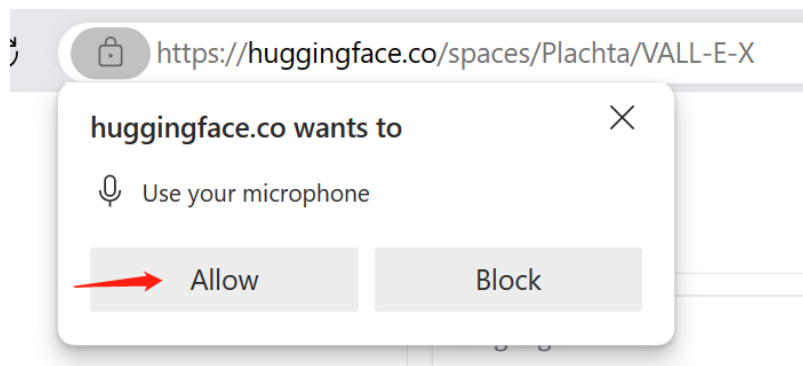
Allow the microphone manually if shows “No microphone found”



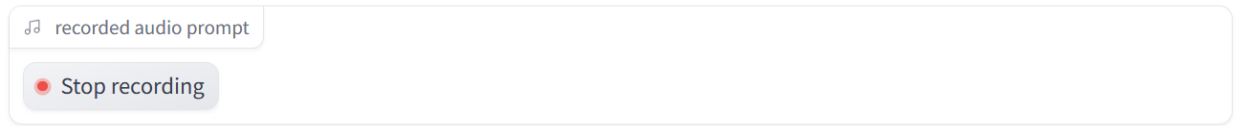
Click the button “Record form microphone”



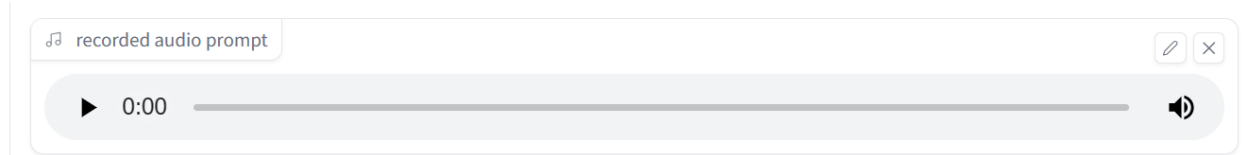
Click “Allow” if you see the prompt as the following:



When you see the following interface, you can start your speaking and click “Stop recording” when you finished.



After recording, your recording file will be presented, you can listen and check the quality of the recording.



2.7 Input the text content you want to synthesis

Text

Welcome back, Master. What can I do for you today?

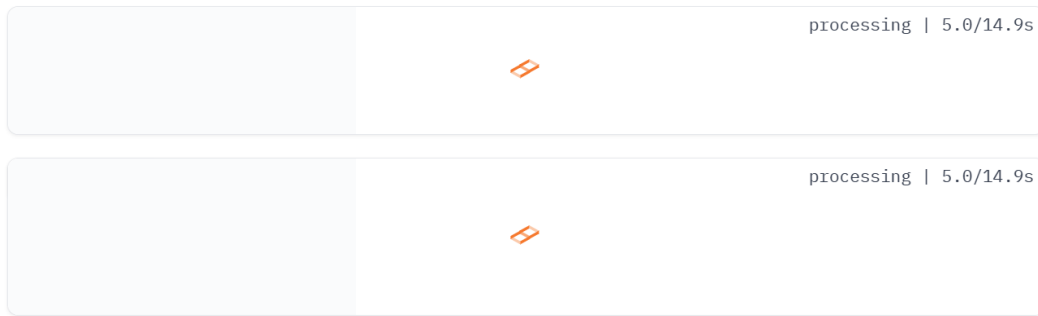
You can modify the content as you like.

2.8 Generate the audio.

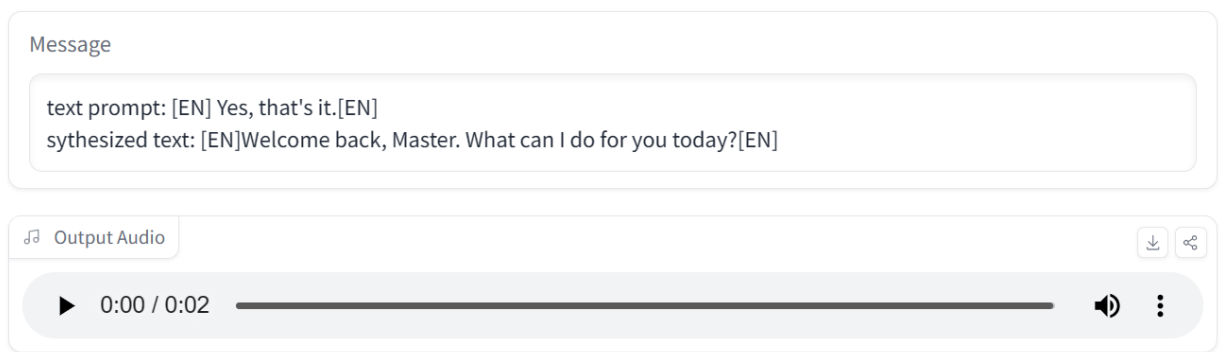
Output Audio

Generate!

- Click the “Generate!” and you can see the generate progress



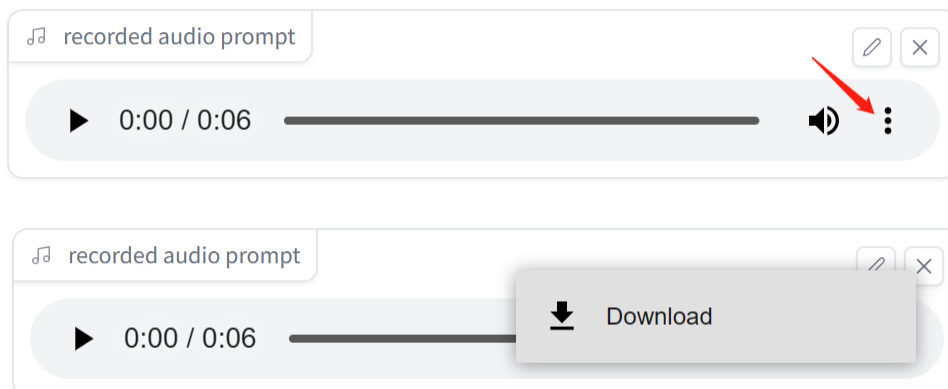
- b. When the generation is finished, you can see the text content recognized from your audio prompt and the generated audio.



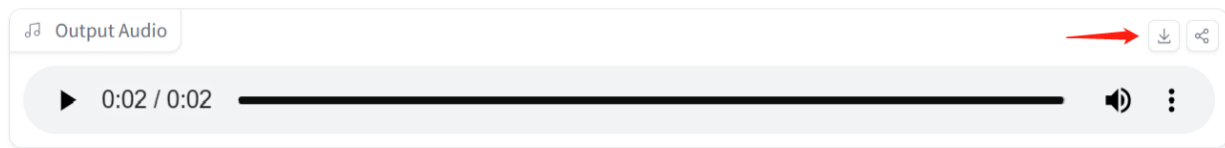
- c. Listen and check the quality of the output audio.

2.9 Download audio prompt / output audio

- Download audio prompt



- Download output audio



2.10 Task

- Please refer to “Lab4_report” for the detail task description and requirements on the submitted report.